Part 1: Diagram 1 – The Commuter's end

Use Case 1: Log In / Sign Up

- **Use Case Name:** Log In / Sign Up
- **Primary Actor:** Registered Commuter
- **Supporting Actor:** Google Authentication Service
- **Goal:** Allow a commuter to securely create an account or access an existing profile using external credential validation.
- **Description:** The commuter initiates authentication. The system offloads the validation handshake to the Google Authentication API. Once verified, the user gains a secure session inside the app.
- **Preconditions:** The application is installed and has a stable active internet connection.
    - The Google Authentication Service API is operational.
- **Trigger:** The commuter opens the app or attempts to execute a protected functional requirement.
- **Main Success Scenario:**
    - The commuter selects the option to Sign In or Sign Up via Google.
    - The system directs the secure authentication frame to the Google Authentication Service API.
    - The commuter provides or selects their active Google credentials.
    - The Google Authentication Service validates the identity and sends a success token back to the system.
    - The system parses the token, establishes a valid user session, and securely maps or logs user identity traits.
    - The system grants full access to the personalized client dashboard.
- **Error Sequences:**
    - **E1: External Service Authentication Failure:** If the Google handshaking token fails or the user cancels the external pop-up frame, the system throws an authorization error message and preserves a logged-out stance.

Use Case 2: View Jeepney Routes & Tricycle Zones

- **Use Case Name:** View Jeepney Routes & Tricycle Zones
- **Primary Actor:** Registered Commuter
- **Supporting Actor:** None
- **Goal:** Allow users to map out, trace, and inspect transit pathways.
- **Description:** The user queries public transit databases to locate geographical representations of tricycle operational sectors and jeepney routes.
- **Preconditions:**
    - <<include>> **Log In / Sign Up:** The user must be authenticated before performing this action.

    - Spatial datasets and path records are populated in the database by the admin backend.
- **Trigger:** The user opens the interactive map layout or searches for a terminal location.
- **Main Success Scenario:**
    - The user requests a route overview or enters a location query.
    - The system verifies active session authentication details (<<include>>).
    - The system parses the geographic input variables.
    - The system queries tricycle zones and jeepney routes from the operational database.
    - The system renders the overlapping jeepney lines and boundary tricycle zones clearly on the viewport interface.
- **Error Sequences:**
    - **E1: No Paths Found:** If the target coordinate parameter exists completely outside documented public transport boundaries, the system suggests nearby standard alternatives.

Use Case 3: Estimate Fares

- **Use Case Name:** Estimate Fares
- **Primary Actor:** Registered Commuter
- **Supporting Actor:** None
- **Goal:** Enable users to calculate estimated transport costs based on origin and destination coordinates.
- **Description:** A commuter inputs waypoints, and the application applies the correct math based on current baseline regulatory fare structures.
- **Preconditions:**
    - <<include>> **Log In / Sign Up:** The user must be authenticated before performing this action.
    - Active fare rule matrices are available in the system repository.
- **Trigger:** The user accesses the fare computation module interface.
- **Main Success Scenario:**
    - The commuter inputs their desired trip origin point and destination.
    - The system verifies active session authentication details (<<include>>).
    - The system calculates the absolute distance along the mapped route segment.
    - The system references base pricing metrics from the active transport database rules.
    - (Step 5) The system computes standard baseline expected transit costs.
    - (Step 6) The system presents the standard total estimate layout to the commuter on screen.
- **Alternative Sequences:**
    - **A1: Profile Adjustments via** <<extend>> **Toggle Accessibility Mode:** If the extension condition is satisfied (the user has flipped the toggle for Accessibility parameters), the system bypasses standard fare calculations and applies specific discount formulas for (Student, PWD, or senior citizen rates).

Use Case 4: Toggle Accessibility Mode

- **Use Case Name:** Toggle Accessibility Mode
- **Primary Actor:** Registered Commuter
- **Supporting Actor:** None
- **Goal:** Enable discounted fare matrices tailored for special groups (students, PWDs, and senior citizens).
- **Description:** This use case **extends** the standard "Estimate Fares" action. It provides custom session rules to calculate discounted outcomes.
- **Preconditions:**
    1. The user is inside the calculator workflow screen.
- **Trigger:** The user voluntarily taps the accessibility overlay options menu.
- **Main Success Scenario:**
    1. The commuter activates the accessibility toggle option on the interface.
    2. The system surfaces profile criteria presets (Student, PWD, Senior Citizen).
    3. The user picks their matching category (Student, PWD, Senior Citizen).
    4. The system updates the evaluation variables within the calculation engine.
    5. The use case automatically **extends** back into the base fare estimation process, returning modified, compliant discounted fare totals.

Use Case 5: Automate Report System for Jeepneys

- **Use Case Name:** Automate Report System for Jeepneys
- **Primary Actor:** Registered Commuter
- **Supporting Actor:** LTFRB Feedback Personnel
- **Goal:** Provide commuters with an easy way to report jeepney related violations and concerns through the official LTFRB complaint channels.
- **Description:** A registered commuter accesses the Report feature to report any issue involving a jeepney, such as overcharging, reckless driving, rude behavior, unsafe driving, refusal to give discount, smoking, vehicle condition, or other service related complaints. The system provides the official LTFRB reporting instructions and offers both a quick online report and the option to file a formal complaint.
- **Preconditions:**
    1. <<include>> **Log In / Sign Up:**
    2. The user must be authenticated before performing this action.
- **Trigger:** The user clicks the Report a Jeepney Concern button.
- **Main Success Scenario:**
    1. The user clicks Report.
    2. The system verifies that the user is logged in.
    3. The user selects Jeepney as the vehicle type.
    4. The system displays the official LTFRB reporting notice.
        - The official LTFRB reporting instructions.
        - A Quick Report (Google Form)
        - A Download Official LTFRB Complaint Form.

5. The user chooses one of the following:
    - Submit a Report (redirects to the official LTFRB Google Form)
    - Download LTFRB Complaint Form (redirects to the official LTFRB Complaint Form for formal complaints).
6. The system redirects the user to the selected official LTFRB reporting page.
7. The user completes the report or downloads the complaint form through the LTFRB website.
8. The system informs the user that they have been redirected to the official LTFRB reporting platform.

Use Case 6: Automated Report for Tricycles

Use Case 7: Help Center

Part 2: Diagram 2 – The Admin's end

Use Case 8: Log In (Admin Gateway)

- **Use Case Name:** Log In (Admin Gateway)
- **Primary Actor:** Admin
- **Supporting Actor:** Google Authentication Service API, System Database
- **Goal:** Enforce a secure operational security gate that validates administrator identity against a pre-whitelisted database records before granting dashboard access.
- **Description:** The system intercepts administrative authentication requests, leveraging the Google Single Sign-On (SSO) framework to extract the user's email token. It then queries the internal database's Admin table to verify if the email is explicitly whitelisted with active administrative clearance.
- **Preconditions:**
    - The admin's specific @gmail.com account has been pre-registered and whitelisted in the system database by a Super Admin.
    - The application is installed and has a stable active internet connection.
    - The Google Authentication Service API is operational.
- **Trigger:** An administrator clicks the administrative login link or attempts to access administrative module endpoints.

Main Success Scenario:

1. The Admin opens the admin control console login interface.

2. The Admin clicks the **"Sign in with Google"** button.
3. The system redirects the user to the Google Authentication interface to select or input their Google account.
4. The Admin successfully completes the Google authentication prompt.
5. The Google Authentication Service securely passes a unique email token back to the backend system.
6. The backend system extracts the Gmail address from the token and executes a lookup query against the database's Admins table.
7. The system successfully verifies that the Gmail address is explicitly pre-whitelisted and possesses active administrative privileges.
8. The backend system removes entry restrictions, initializes a secure administrative session token, and loads the management workspace dashboard.

Use Case 9: User Management

- **Use Case Name:** User Management
- **Primary Actor:** Admin
- **Supporting Actor:** None
- **Goal:** Maintain total control over active profiles, blocking or managing accounts to safeguard user data integrity.
- **Description:** Administrators read, inspect, audit, and modify profile states across the active user account database.
- **Preconditions:**
    1. <<include>> **Log In / Sign Up:** The Admin must be authenticated before performing this action.
- **Trigger:** The Admin launches the account security audit console.
- **Main Success Scenario:**
    1. The Admin selects the "User Management Profile Dashboard".
    2. The system verifies active administrative authentication details (<<include>>).
    3. The system generates a searchable catalog representation containing active commuter accounts.
    4. The Admin tracks account behaviors, reads logs, or takes control actions (e.g., updates metadata details, flags, or suspends accounts violating safety terms).
    5. The system records administrative configuration changes instantly inside the user profile records.

Use Case 10: Manage Route Data

- **Use Case Name:** Manage Route Data
- **Primary Actor:** Admin
- **Supporting Actor:** None
- **Goal:** Maintain accurate definitions of active public utility paths, route legs, and service lines.

- **Description:** The Admin alters, appends, updates, or purges structural jeepney routes and tricycle sectors within the app.
- **Preconditions:**
    1. <<include>> **Log In / Sign Up:** The Admin must be authenticated before performing this action.
- **Trigger:** Regional route revisions, transport terminal reallocations, or physical infrastructure updates occur.
- **Main Success Scenario:**
    1. The Admin clicks on the route data configuration module interface.
    2. The system verifies active administrative authentication details (<<include>>).
    3. The system pulls current mappings and route parameters onto the screen.
    4. The Admin executes core database operations: Adds a new route, edits an existing coordinate segment, or deletes a redundant path option.
    5. The Admin commits the modification to production.
    6. The database saves and propagates changes immediately to all front-end mobile clients.

Use Case 11: Update Fare Matrices

- **Use Case Name:** Update Fare Matrices
- **Primary Actor:** Admin
- **Supporting Actor:** None
- **Goal:** Keep automated calculation algorithms structurally aligned with current laws.
- **Description:** The Admin edits base vehicle values and distance pricing multipliers whenever official regulatory bodies update regional transport rates.
- **Preconditions:**
    1. <<include>> **Log In / Sign Up:** The Admin must be authenticated before performing this action.
- **Trigger:** An official legal announcement concerning regional price adjustments is mandated.
- **Main Success Scenario:**
    1. The Admin inputs changes via the "Pricing Configurations Console Matrix".
    2. The system verifies active administrative authentication details (<<include>>).
    3. The Admin types new rates for base drop fees or incremental per-kilometer rates across target vehicle categories.
    4. The Admin confirms and publishes changes.
    5. The system applies the modifications to the math engine, shifting the pricing logic globally across user devices.

Use Case 12: View Analytics

- **Use Case Name:** View Analytics
- **Primary Actor:** Admin
- **Supporting Actor:** None

- **Goal:** Pull and extract commuter interaction data for independent external analysis.
- **Description:** The Admin navigates to the backend analytics utility. The system aggregates the raw data and packages it into a structured, downloadable file format.
- **Preconditions:**
    1. <<include>> **Log In / Sign Up:** The Admin must be authenticated before performing this action.
- **Trigger:** The Admin accesses the data extraction panel to compile data for external tools (like Excel, PowerBI, Jupyter Notebook).
- **Main Success Scenario:**
    1. The Admin navigates to the "Raw Analytics Export" utility section.
    2. The system verifies active admin authentication details (<<include>>).
    3. The system presents raw data on the interface.
    4. The Admin configures their desired dataset and clicks the "Extract Raw Dataset" button.
    5. The system automatically packages the unformatted data into a downloadable file extension (.csv or .xlsx).
    6. The system triggers a file download prompt on the Admin's browser or device screen.
    7. The Admin successfully saves the raw data file locally to begin independent analysis.

Non-Functional Requirements

The app should be easy to use, fast, reliable, secure, and maintainable. For a commuter app, the most important non-functional requirements are usually usability, performance, reliability, and supportability.

- **Usability:** The interface should be simple enough for commuters to search routes quickly on a phone screen.
- **Performance:** Search results should load within an acceptable time even under normal mobile network conditions.
- **Reliability:** The app should handle failed requests, missing data, and poor connectivity without crashing.
- **Security:** User accounts, saved preferences, and admin functions should be protected from unauthorized access.
- **Maintainability:** Admin tools should make it easy to update routes, fares, or trip data without changing the whole system.
- **Compatibility:** The app should run properly on the supported Android and iOS devices.